

Supporting Adult Guidance

Estimation · Protecting judgement, agency and a calm return to prerequisite ideas

Connection before curriculum. Always.

The adult's role is to protect thinking and regulation. Invite estimation before exact calculation, and use a Reconnection Route only when a specific prerequisite is materially blocking progress.

Lesson purpose at a glance

The learner rounds values to friendly numbers, estimates calculations and uses the estimate to judge whether an exact or calculator result is reasonable.

What the learner is developing

- magnitude and place-value awareness;
- rounding to 1 significant figure for a purpose;
- mental calculation with friendly values;
- reasonableness and error detection;
- confidence to question a displayed answer.

Key mathematical idea

Estimation preserves scale rather than exact detail. For 48.7×19.6 , the friendly calculation $50 \times 20 = 1,000$ predicts the magnitude. The exact result 954.52 is therefore plausible; 95.452 and 9,545.2 are suspicious.

Preparation and Reconnection Routes

Essential preparation

- Open the lesson and confirm that Guidance, Reconnection Routes, Scratchpad and Practice Companion controls are visible.
- Have paper or another recording option available.
- Check that the learner knows they may pause or continue later.

Reconnection is attached to need

Reconnection is not a separate catch-up block. Use the in-page button only when place value, rounding, magnitude or an operation is preventing the learner from entering the current estimation task.

1. Name the specific blocking idea neutrally.
2. Invite the shortest relevant route.
3. Allow the learner to answer, sketch or use a hint.
4. Record only the minimum return evidence needed.
5. Use Return to Estimation and resume at the original task.

Useful language

“Would a short place-value route help here?” “You can return to this calculation as soon as that one connection feels secure enough.”

Supporting the six Cornerstones

Connection

Ask: “Without calculating exactly, which answer could be sensible?”

Movement

Invite the learner to change one value and predict how the estimate moves.

Reflection

Ask whether the estimate correctly predicted the scale and whether it was close enough for the purpose.

Creativity

Invite a calculation with a plausible and a deliberately suspicious answer.

Rest

Use the calm retrieval route: round → calculate → check.

Nutrition

Name intellectual nourishment honestly: magnitude, judgement, efficiency and confidence.

Invitational rhythm

Connect 8 · Move 18 · Practise 10 · Reflect/Create 5 · Rest/Nourish 2 · Evidence 2. These are orientation points, not deadlines.

Common difficulties and supportive responses

The learner calculates exactly first.

Ask for the likely size before checking exact digits.

The learner rounds every value to the same number of decimal places.

Return to the first non-zero digit and discuss 1 significant figure as a useful friendly-number strategy.

A decimal product is ten or one hundred times too large.

Estimate with 0.5×60 or a similar friendly calculation, then reconsider decimal placement.

The estimate is treated as wrong because it differs from the exact answer.

Clarify that the estimate has a different job: scale and reasonableness, not exact precision.

Division feels difficult.

Look for compatible friendly values, such as $80 \div 4$.

The learner appears overloaded.

Reduce the visible task, offer another recording mode, or pause without converting the pause into failure.

Practice Companion support pathway

Universal worked-solution release rule

For every question independently: Socratic prompt used + Hint used + two non-empty answers entered and checked = Worked solution released.

6. Allow time to think, sketch or begin independently.
7. Invite the Socratic prompt if useful.
8. Let the learner enter and check a first answer.
9. Invite the Hint if useful.
10. Let the learner enter and check a second answer.
11. Only then may the Worked solution unlock for that question.

Do not provide the complete worked solution verbally while the button is locked. You may clarify vocabulary, read the question aloud, support regulation or invite a Reconnection Route without supplying the method.

QA observation

- Prompt, hint and attempts reset independently for each question.
- An empty answer does not count as an attempt.
- Navigation must retain the correct state for each question.
- Equivalent correct numerical forms should be accepted.

Access, evidence and a valid finish

Access and regulation

- Use age-neutral language and avoid deficit framing.
- Allow spoken, typed, drawn, calculated or annotated responses.
- Keep the Scratchpad optional and accept paper or another tool.
- Do not use time-pressure language.
- Treat incomplete work as information for the next connection.

Evidence to notice

- Does the learner predict the likely magnitude?
- Can they select friendly values?
- Can they distinguish estimate from exact answer?
- Can they use estimation to challenge a suspicious display?
- Can they explain whether the estimate was useful?

A valid finishing point

The learner may finish after one secure estimation decision, one clear explanation or one identified Reconnection Route and return point. The next lesson decision should follow the evidence and regulation state, not a fixed completion demand.

Observation notes

Secure connection noticed: _____

Reconnection Route used and return evidence:

Next mathematical connection:

Access or regulation support that helped:
